

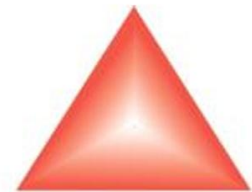


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المملكة المغربية
وزارة التعليم العالي
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Doctoral Studies Center in Information and Engineering Sciences and Technologies (ST2I)

Explainable Artificial Intelligence (XAI) for Decision-Making in Graph-Based Recommendation Systems

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Introduction

Ubiquity

How do recommender systems shape what billions of users see, buy, and watch every day?

The Black Box

Why do state-of-the-art graph based RS remain opaque to the users and designers who rely on them?

Toward Trust

How can transparency be built as part of the recommendation process instead of afterwards?

Motivating references

- [1] Y. Zhang and X. Chen, "Explainable Recommendation: A Survey and New Perspectives," *Foundations and Trends in Information Retrieval*, vol. 14, no. 1, pp. 1–101, 2020.
- [2] D. Gunning and D. W. Aha, "DARPA's Explainable Artificial Intelligence (XAI) Program," *AI Magazine*, vol. 40, no. 2, pp. 44–58, 2019.
- [3] A. Das and P. Rad, "Opportunities and Challenges in Explainable Artificial Intelligence (XAI): A Survey," arXiv:2006.11371, 2020.
- [4] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, "Explainable AI: A Review of Machine Learning Interpretability Methods," *Entropy*, vol. 23, no. 1, p. 18, 2021.
- [5] J. Kunkel, T. Donkers, L. Michael, C.-M. Barbu, and J. Ziegler, "Let Me Explain: Impact of Personal and Impersonal Explanations on Trust in Recommender Systems," in *Proc. CHI Conf. Human Factors in Computing Systems (CHI)*, 2019, pp. 1–12.
- [6] A. Jacovi, A. Marasović, T. Miller, and Y. Goldberg, "Formalizing Trust in Artificial Intelligence: Prerequisites, Causes and Goals of Human Trust in AI," in *Proc. ACM Conf. Fairness, Accountability, and Transparency (FAccT)*, 2021, pp. 624–635.

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General Context

Examples of AI Powered Recommendation systems



Netflix – Movie and TV recommendation platform.



Spotify – Music recommendation and streaming service.



Yelp – Local business and restaurant recommendation platform.



Amazon – E-commerce platform with personalized product recommendations.

[7] S. Raza, M. Rahman, S. Kamawal, A. Toroghi, A. Raval, F. Navah, and A. Kazemeini, "A Comprehensive Review of Recommender Systems: Transitioning from Theory to Practice," Computer Science Review, 2026.

[8] O. A. S. Ibrahim, E. M. G. Younis, E. A. Mohamed, and W. N. Ismail, "Revisiting Recommender Systems: An Investigative Survey," Neural Computing and Applications, 2025.

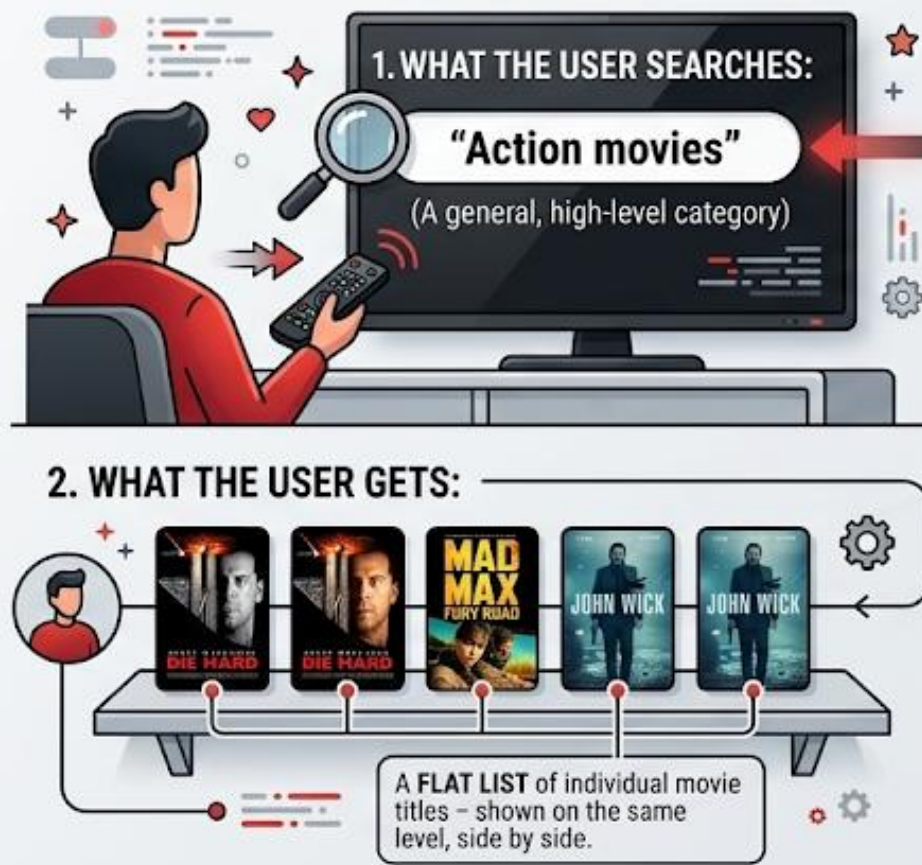
[9] L. Harichandan, S. Nayak, and R. K. Lenka, "A Comprehensive Review on Video Recommendation Systems: Models, Challenges, and Applications," Journal of Engineering Science and Technology Review, 2025.

[10] S. Raza, M. Rahman, S. Kamawal, A. Toroghi, A. Raval, F. Navah, and A. Kazemeini, "A Comprehensive Review of Recommender Systems: Transitioning from Theory to Practice," arXiv preprint arXiv:2407.13699, 2024.

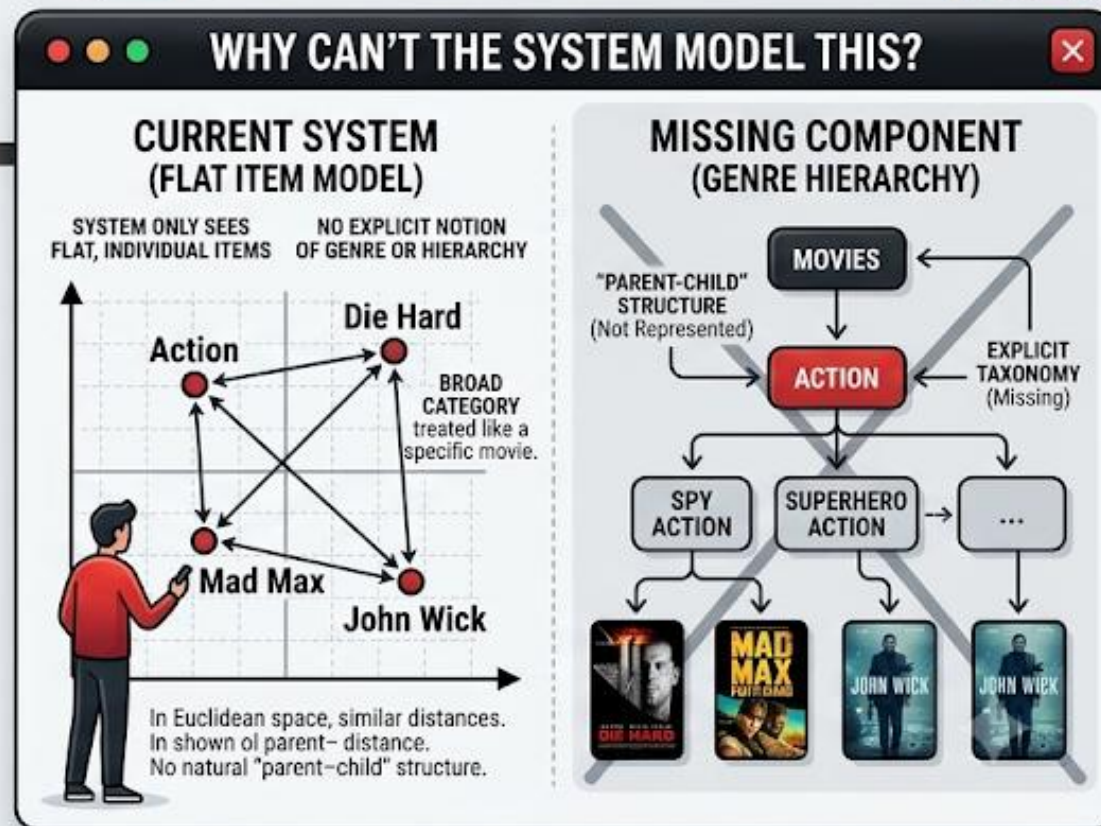
General Context

Examples of AI Powered Recommendation systems

The User Action and Results

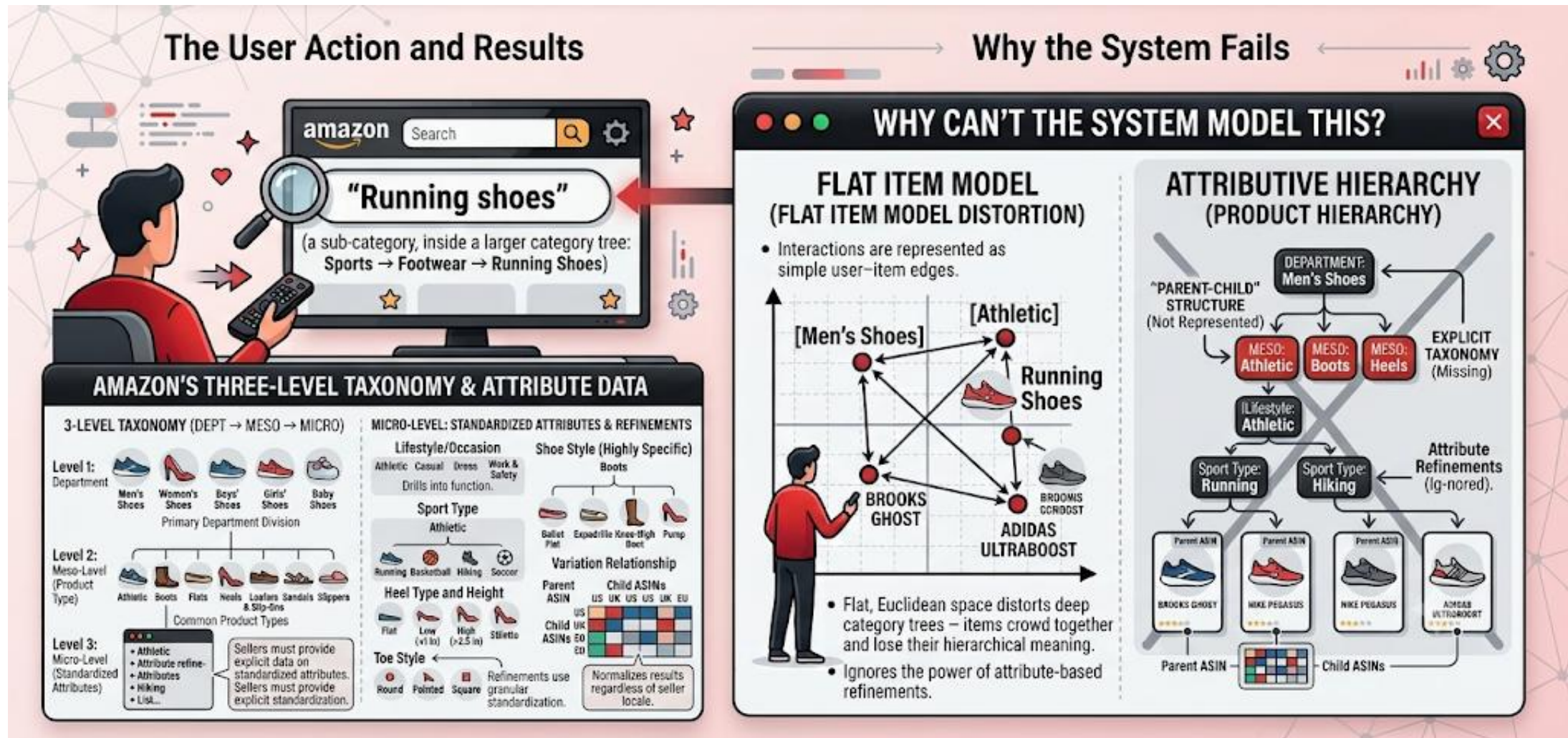


Why the System Fails



General Context

Examples of AI Powered Recommendation systems



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Context and Problematic

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- Recommendations Systems promotes products by leveraging the internet and electronic devices for efficient global reach.
- It offers new ways to communicate, engage and convert audiences into customers.



Predictive Analytics



Automation



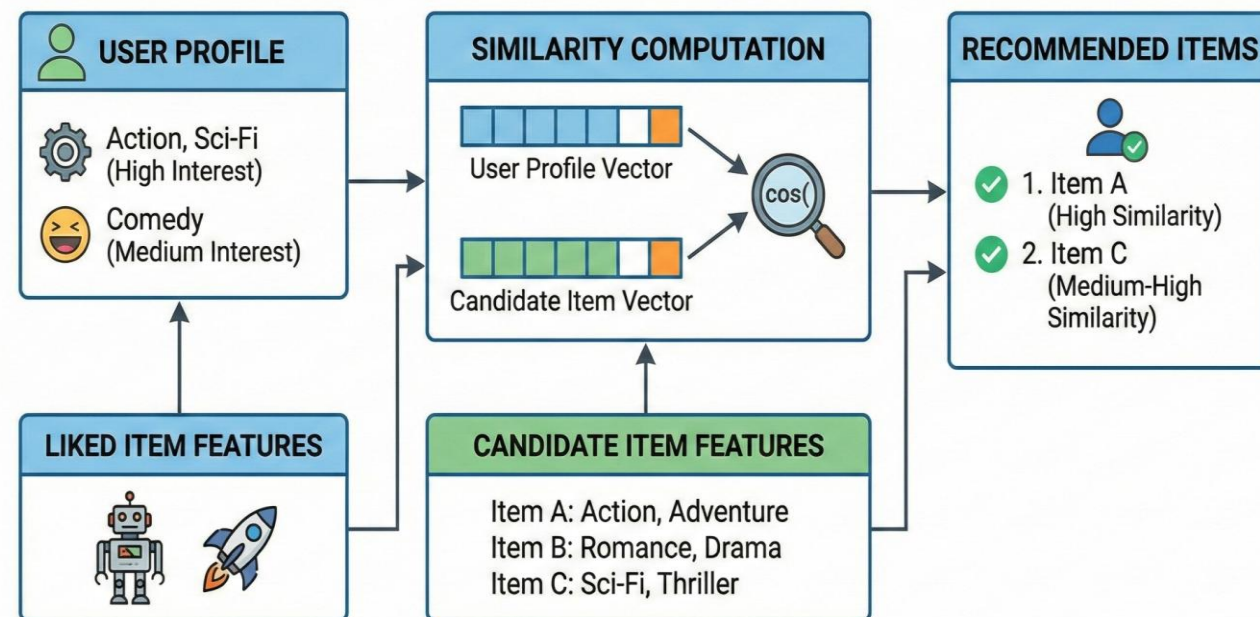
Data Insights



Enhanced Decision Making

Content Based Filtering

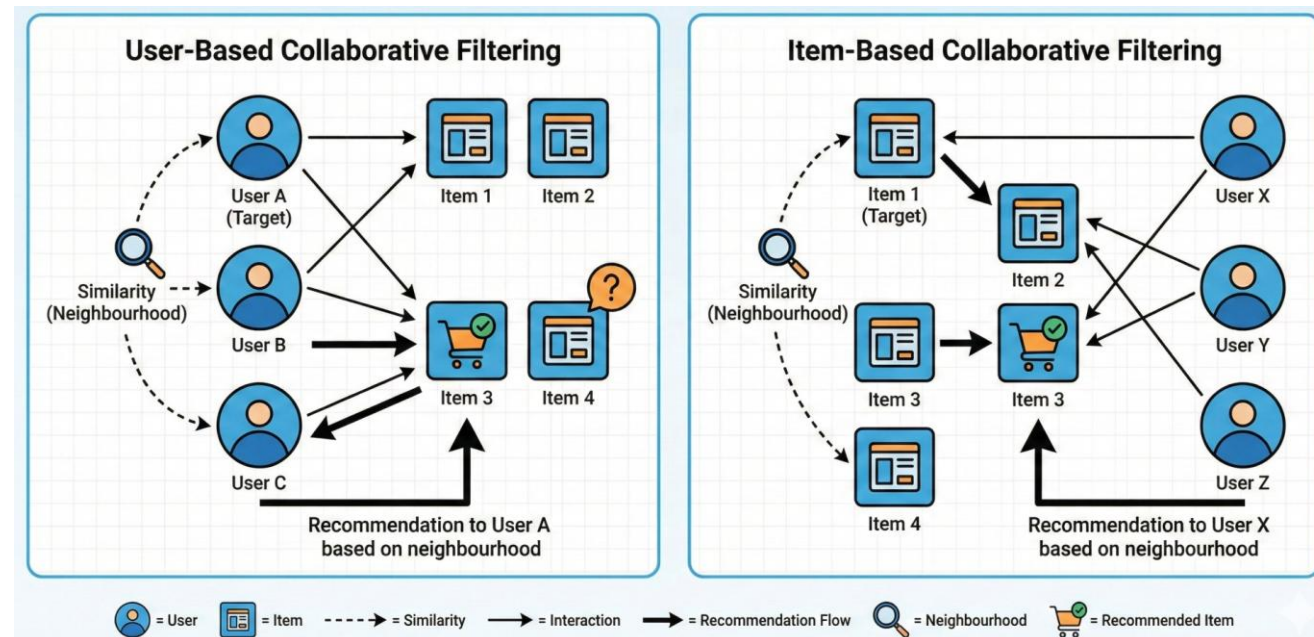
- Recommends items based on their features and the user's past preferences.
- Builds a user profile from previously liked, viewed, or selected items.
- Suggests new items with similar characteristics.
- Limitation: recommendations may become too similar and lack diversity.



[11] M. J. Pazzani and D. Billsus, "Content-Based Recommendation Systems," in The Adaptive Web, P. Brusilovsky, A. Kobsa, and W. Nejdl, Eds. Berlin, Germany: Springer, 2007, pp. 325–341.

Collaborative Filtering

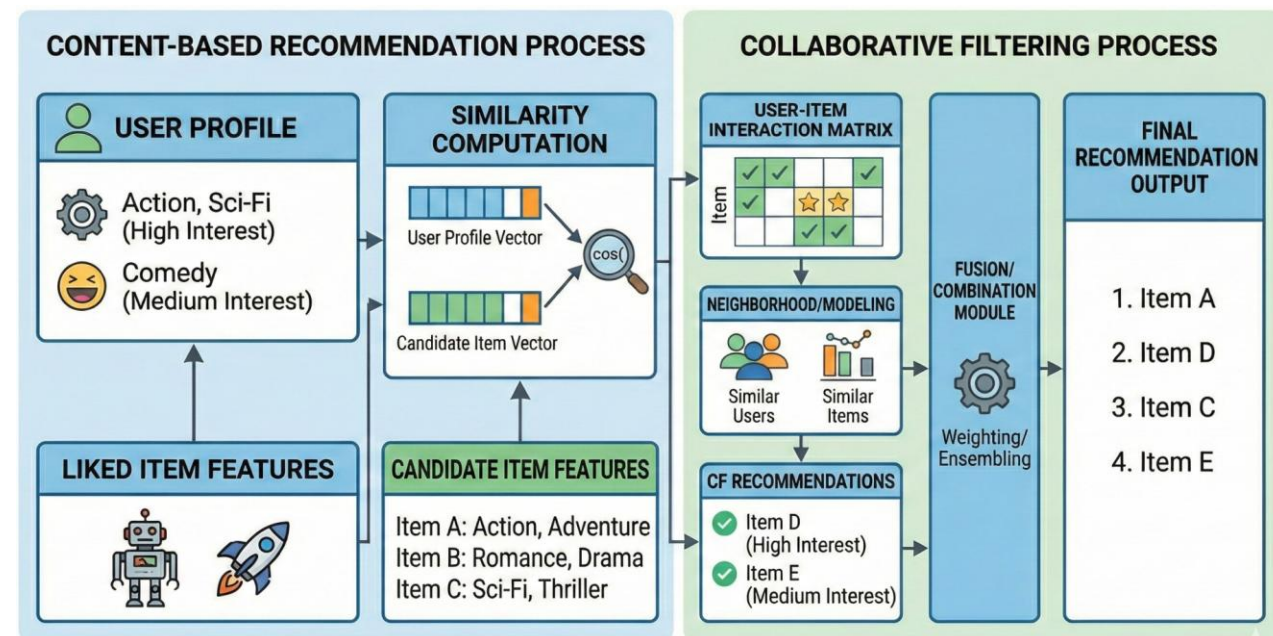
- Recommends items using preferences and behaviors of similar users.
- Identifies patterns in user-item interactions, such as ratings, clicks, or purchases.
- Can be user-based or item-based collaborative filtering.
- Limitation: it struggles with new users or new items due to limited data.



[12] J. L. Herlocker, J. A. Konstan, A. Borchers, and J. Riedl, "An Algorithmic Framework for Performing Collaborative Filtering," in Proc. 22nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Berkeley, CA, USA, 1999, pp. 230–237.

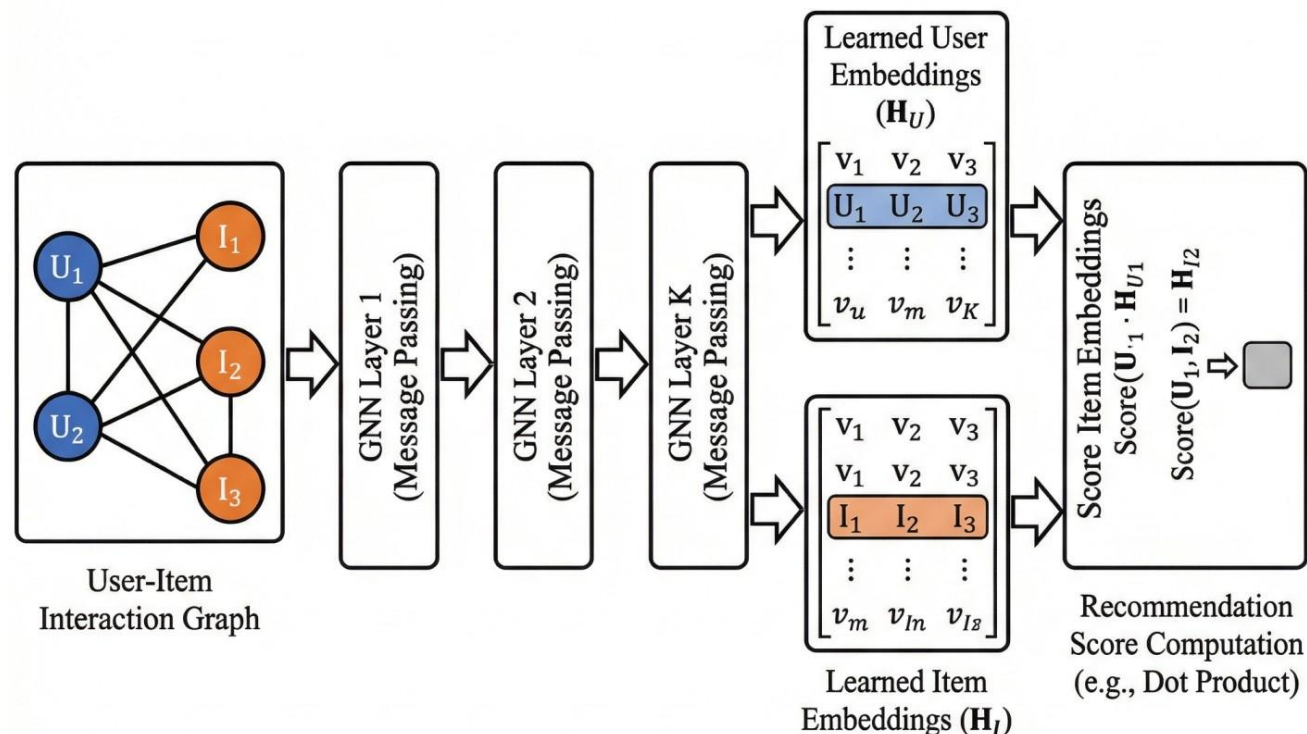
Hybrid Recommendation Approach

- Combines content-based and collaborative filtering methods.
- Uses both item features and user-item interaction data.
- Improves recommendation accuracy and reduces the weaknesses of individual methods.
- Can better handle issues such as limited data, overspecialization, and the cold-start problem.



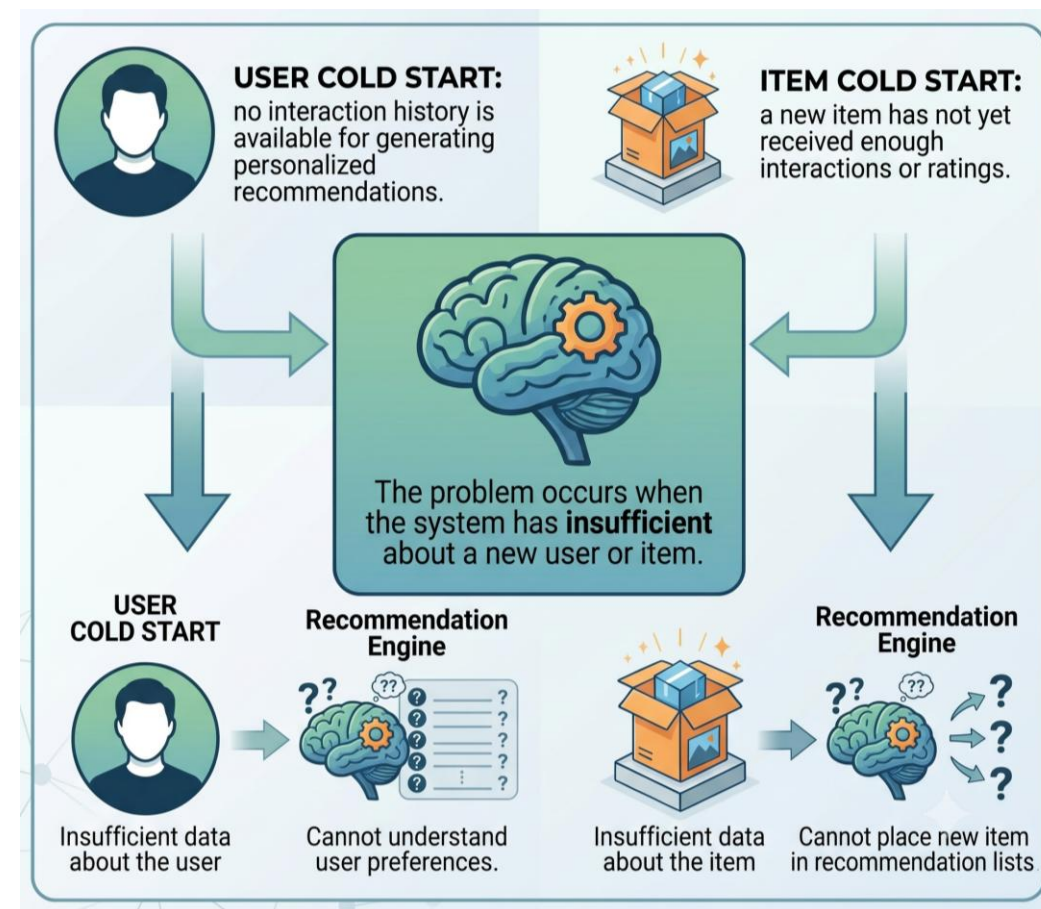
Graph Based Recommendation Systems

- Represents users, items, and their interactions as nodes and edges in a graph.
- Uses graph structure to discover relationships and multi-hop connections.
- Applies methods such as random walks, graph embeddings, or graph neural networks.
- Limitation: large graphs can require significant computation and memory.



Cold Start-Problem

- Occurs when the system has insufficient data about a new user or item.
- User cold start: no interaction history is available for generating personalized recommendations.
- Item cold start: a new item has not yet received enough interactions or ratings.
- Common solutions include onboarding questions, demographic data, item features, and hybrid recommendation methods.



[15] M. Kolli, L. D. Mukil, N. Kartthikeyan, M. Pranavkrishnan, N. Sampath, and P. C. Nair, "Addressing the Cold Start Challenge through the Implementation of Content-Based Filtering and Hybrid Filtering for Movie Recommendation," in Proc. 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), IEEE, 2024.

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Context and Problematic

Recommendation Systems

Problematic & Objectives

Contributions

Graph-based recommender systems deliver strong performance, but they often remain black boxes that are difficult to explain.

Existing explainability methods are often post hoc and weakly integrated into the recommendation process [11].

Design a unified framework combining accuracy, explainability, adaptability, and hierarchical modeling.

User preferences evolve over time, but most models struggle to adapt to these changes [12].

Hierarchical structures, categories, subcategories, and taxonomies are poorly represented in conventional Euclidean spaces.

[11] A. Kamal, A. Ragno, M. Plantevit, et al., "Leveraging Internal Representations of GNNs With Shapley Values," *Data Min. Knowl. Discov.*, vol. 39, no. 89, 2025.

[12] L. Wu, D. Hao, and Z. Fan, "Explainable Graph Neural Networks via Structural Externalities," in *Proc. 34th Int. Joint Conf. Artif. Intell. (IJCAI)*, 2025, pp. 583–591.

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Context and Problematic

Recommendation Systems

Problematic & Objectives

Contributions

Graph-based collaborative filtering with a LightGCN style recommender

Shapley-value-based explanation of historical user-item interactions

Adaptive recommendation through contribution-aware inference-time reweighting

Hierarchical recommendation using hyperbolic graph neural networks

Unified experimental validation on MovieLens and Amazon, combining accuracy and interpretability analysis



Protocols

MovieLens 100k

- 100,000 ratings from about 1,000 users on 1,700 movies, collected between 1997 and 1998.
- One of the earliest standard recommender benchmarks, still used today for fast prototyping and ablation studies
- Its smaller size lets me run ablations and sensitivity tests quickly, alongside the larger MovieLens 1M.

Specification	MovieLens 100K
Number of users	943
Number of items	1,682 movies
Number of interactions	100,000 ratings
Rating scale	1–5 stars
Minimum interactions per user	20 ratings
Approximate sparsity	93.7%

[16] F. Maxwell Harper and J. A. Konstan, "MovieLens 100K Dataset," GroupLens Research, University of Minnesota. Available: <https://grouplens.org/datasets/movielens/100k/>

[17] F. M. Harper and J. A. Konstan, "The MovieLens Datasets: History and Context," ACM Transactions on Interactive Intelligent Systems, vol. 5, no. 4, pp. 1–19, Dec. 2015, doi: 10.1145/2827872

[18] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, "Recommender Systems Survey," Knowledge-Based Systems, vol. 46, pp. 109–132, Jul. 2013

MovieLens 100k

- **Recommendation Performance** Provides user–movie rating data to train and evaluate recommendation algorithms.
- **Algorithm Benchmarking** Serves as a standard benchmark for fair comparison of collaborative filtering, graph neural networks, and deep learning methods.
- **Explainable Recommendation** Enables analysis of user–item interactions to generate and evaluate interpretable recommendations.

User ID	Movie ID	Movie Title	Rating	Timestamp
1	1	Toy Story (1995)	5	874965758
1	32	Crumb (1994)	3	876893171
2	257	Men in Black (1997)	4	888550871
5	474	Dr. Strangelove (1964)	5	875635845
10	50	Star Wars (1977)	4	877888877

MovieLens 1M

- About 1 million ratings, from 6,000 users on 4,000 movies, collected by GroupLens Research in 2000.
- The standard benchmark for collaborative filtering research for over two decades, including LightGCN, NGCF, and most GNN recommenders.
- Using this benchmark keeps my results directly comparable with prior work in graph-based recommendation.

Specification	MovieLens 1M
Number of users	6,040
Number of items	3,952 movie
Number of interactions	1,000,209 ratings
Rating scale	1–5 stars
Minimum interactions per user	20 ratings
Approximate sparsity	95.8%

[19] GroupLens Research, "MovieLens 1M Dataset," University of Minnesota. Available: <https://grouplens.org/datasets/movielens/1m/>

[20] F. M. Harper and J. A. Konstan, "The MovieLens Datasets: History and Context," ACM Transactions on Interactive Intelligent Systems, vol. 5, no. 4, Art. no. 19, Dec. 2015, doi: 10.1145/2827872.

[21] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, "Recommender Systems Survey," Knowledge-Based Systems, vol. 46, pp. 109–132, Jul. 2013.

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Experiment Setup

Datasets

Metrics

Hardware

MovieLens 1M

- Scalable Recommendation Evaluation: Provides over 1 million user–movie ratings, enabling the evaluation of recommendation algorithms on a larger and more realistic dataset than MovieLens 100K.
- Benchmarking Advanced Models: Serves as a standard benchmark for comparing collaborative filtering, matrix factorization, graph neural networks, deep learning, and explainable recommendation methods.
- Learning Rich User Preferences: Includes demographic information (age, gender, occupation), movie genres, timestamps, and ratings, allowing researchers to model user behavior and improve recommendation quality.

MovieLens 1M

User ID	Movie ID	Movie Title	Rating	Gender	Age	Occupation	Timestamp
1	1193	One Flew Over the Cuckoo's Nest	5	F	16	10	978300760
1	661	James and the Giant Peach	3	F	17	10	978302109
2	914	My Fair Lady	4	M	56	16	978298709
5	3408	Erin Brockovich	4	M	25	20	978245234
10	2571	Matrix, The	5	F	35	1	978228345

Amazon Products

- Large-scale e-commerce ratings and reviews collected by the McAuley Lab, UC San Diego, through 2018.
- Review-based and sentiment-aware recommendation, and evaluating models at realistic, large, sparse scale
- It tests whether my methods generalise beyond movies, to e-commerce data with very different scale and sparsity.

Specification	Amazon Products
Number of users	≈15.5 million
Number of items	≈9.4 million products
Number of interactions	≈233.1 million reviews
Rating scale	1–5 stars
Minimum interactions per user	No fixed minimum
Approximate sparsity	Extremely sparse

[22] J. McAuley Lab, "Amazon Review Data (2018)," University of California San Diego. [Online]. Available: https://cseweb.ucsd.edu/~jmcauley/datasets/amazon_v2/

Amazon Products

- **Large-Scale Recommendation:** Provides millions of user–product interactions for training and evaluating recommendation algorithms on real-world e-commerce data.
- **Personalized Product Recommendation:** Includes user ratings, product metadata, categories, brands, and review history to model user preferences and improve recommendation quality.
- **Explainable and Context-Aware Recommendation:** Contains review text, helpfulness votes, and product attributes, enabling explainable recommendations and sentiment-aware recommendation models.

Amazon Products

User ID	Product ID (ASIN)	Category	Rating	Review Summary	Brand
A1B2C3	B000123XYZ	Electronics	5	Excellent sound quality	Sony
A4D5E6	B000456ABC	Books	4	Informative and well written	Penguin
A7F8G9	B000789DEF	Home & Kitchen	3	Average quality	KitchenAid
A2H3I4	B000321GHI	Clothing	5	Fits perfectly	Levi's
A5J6K7	B000654JKL	Sports	4	Good value for money	Nike

• Metric	• Formula	• Why used
• Precision	• $Precision = \frac{TP}{(TP+FP)}$	• Measures how many predicted links/interactions are correct.
• Recall	• $Recall = \frac{TP}{(TP+FN)}$	• Measures how many real links/interactions the model successfully recovers.
• Precision@K	• $Precision@K = \frac{1}{k} \sum_{i=1}^k rel_i$	• Measures the proportion of relevant items among the top-K recommendations.
• Recall@K	• $Recall@K = \frac{\frac{1}{k} \sum_{i=1}^k rel_i}{ \mathcal{R}_U }$	• Measures how many of the user's relevant items are successfully retrieved in the top-K recommendations.
• F1-Score@K	• $F1@K = 2 \times \frac{Precision@K \times Recall@K}{Precision@K + Recall@K}$	• Provides a balanced score between precision and recall.

Notations

TP = true positives, FP = false positives, FN = false negatives, rel_i = relevance of item at rank i , and $\mathcal{R}_U$ = total number of relevant items for user u .

Contribution	Hardware	CPU / Chip	RAM	Storage
First	Lenovo ThinkPad X1 Carbon	Intel Core i7-7600U	16 GB	512 GB
Second and Third	MacBook Pro	Apple M4 Max	48 GB	1 TB

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Contribution I

Collaborative Filtering Movies Recommendation System based on Graph Neural Networks

Research Gap

- Sparse interaction matrices provide limited learning signals.
- Classical collaborative filtering poorly captures indirect, multi-hop user–item relationships.
- Matrix-based models ignore the natural graph structure of recommendation data.

Research Gap

There is a need for a graph-based collaborative filtering model that directly exploits higher-order user–item connectivity, without relying on side information, to improve recommendation performance in sparse settings.

[23] Y. Zhang and X. Chen, "Explainable Recommendation: A Survey and New Perspectives," IEEE Access, vol. 8, pp. 110–120, 2020.

[24] B. P. Chamberlain, S. R. Hardwick, D. R. Wardrop, F. Ozogang, F. Daolio, and S. Vargas, "Scalable Hyperbolic Recommender Systems," arXiv:1902.08648, 2019.

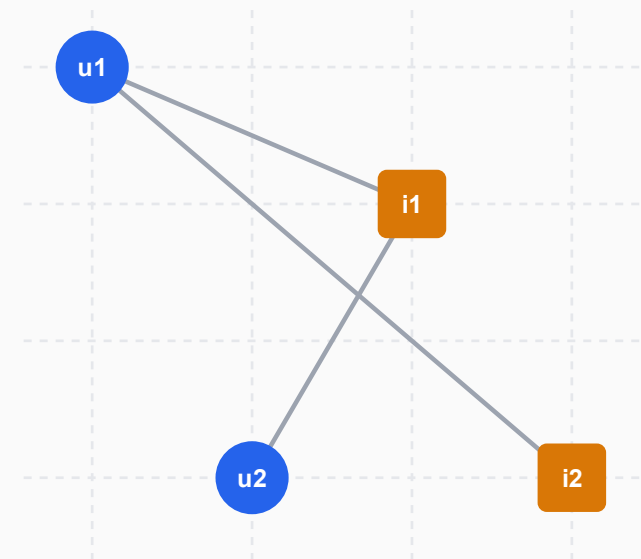
Research Questions → Objectives

ID	Research Question	Objective
RQ1.1	Does graph-based link prediction outperform traditional collaborative filtering methods?	Compare LightGCN with UBCF, IBCF, MF, SVD, and SVD++.
RQ1.2	Do multi-hop propagation and layer aggregation improve collaborative learning?	Combine user-item representations across multiple graph layers.
RQ1.3	Can a simple graph model support stable future extensions?	Use linear aggregation and inner-product scoring.
RQ1.4	What limitations remain despite improved accuracy?	Identify static, uniform, and opaque interactions to motivate Contribution 2.

Euclidean Space

- Flat, normal geometry distance is a straight line between two points, the space we use every day.
- The standard choice for embeddings: matrix factorisation, word embeddings, and most graph neural networks, including LightGCN.
- It is the natural starting point and the space used in Contribution I, before hierarchy motivates a different geometry later.

A. Bipartite Graph & Matrix Space

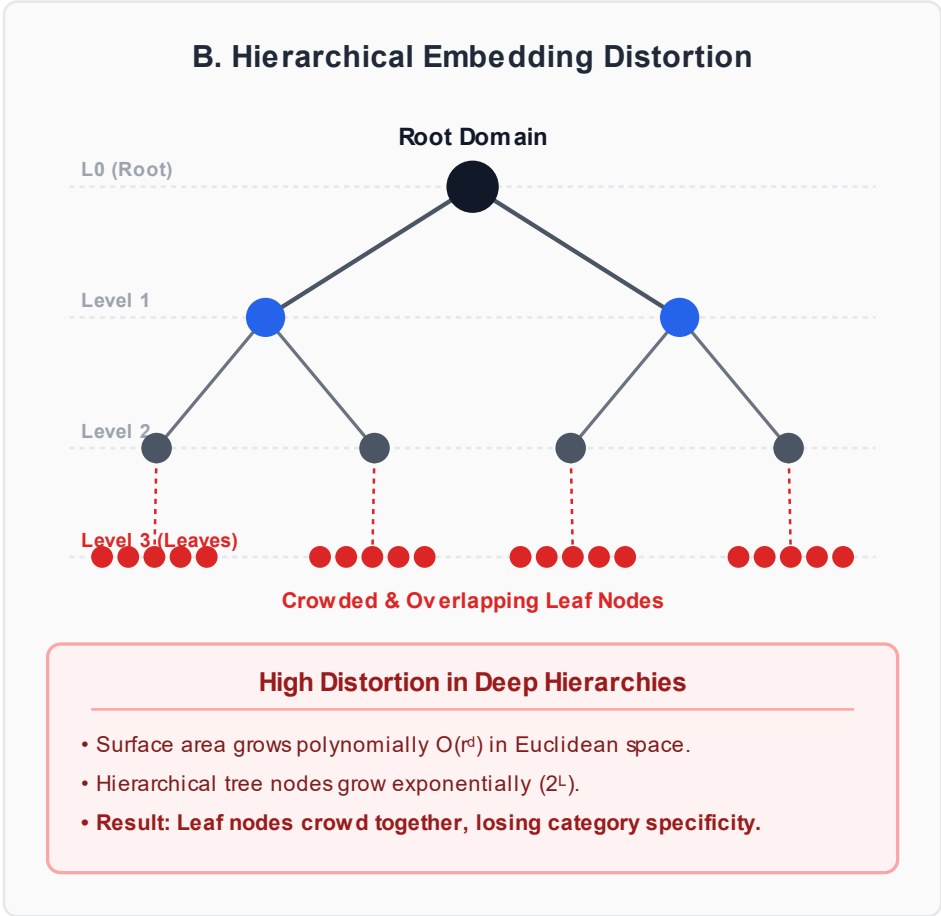
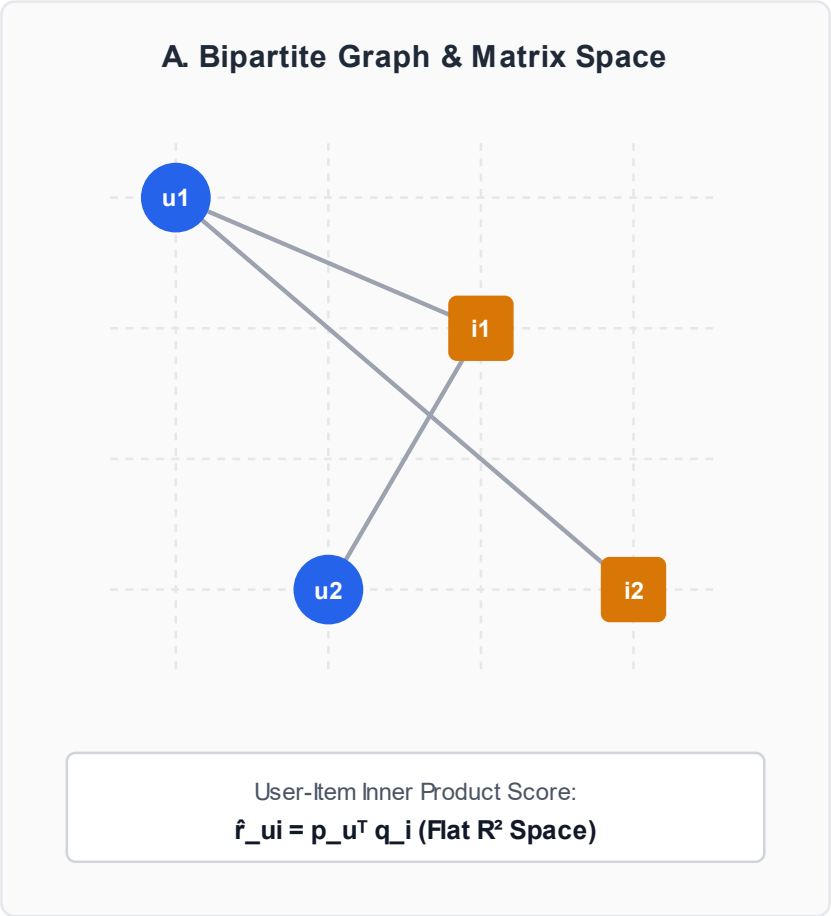


User-Item Inner Product Score:

$$\hat{r}_{ui} = \mathbf{p}_u^T \mathbf{q}_i \text{ (Flat } \mathbb{R}^2 \text{ Space)}$$

[25] Y. Koren, R. Bell, and C. Volinsky, "Matrix Factorization Techniques for Recommender Systems," IEEE Computer, vol. 42, no. 8, pp. 30–37, 2009.

Euclidean Space



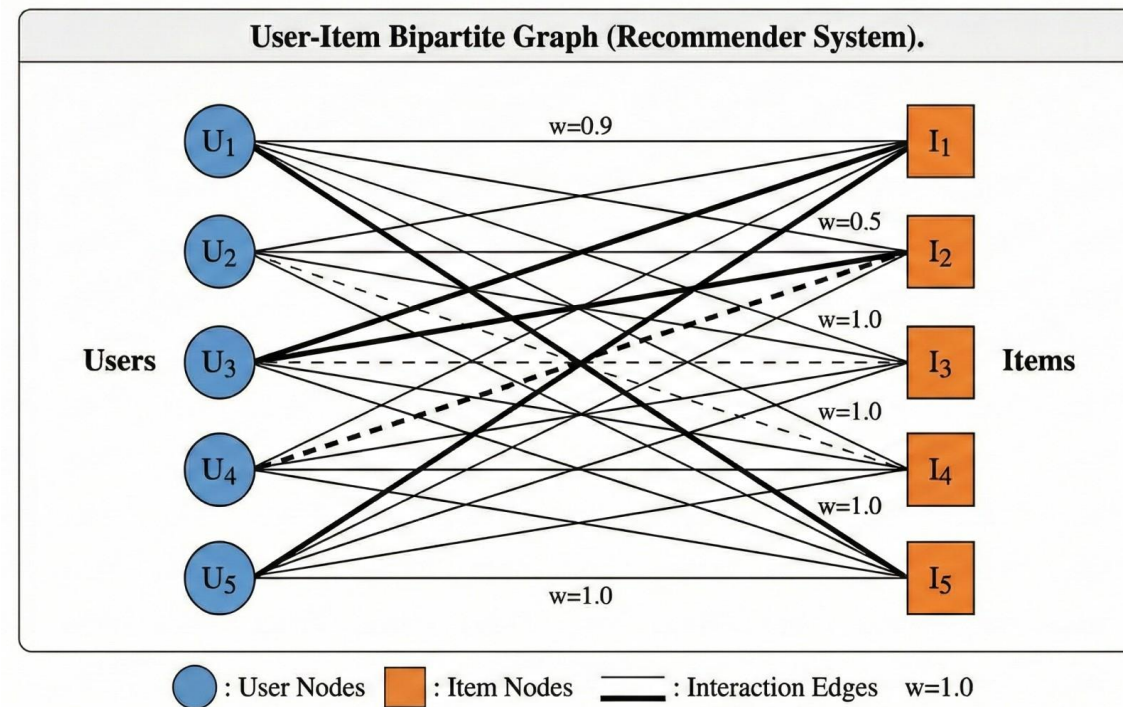
Architecture Walkthrough

Step1

1.User-Item Interaction Input

2.Bipartite Graph Construction

3.Embedding Initialization



Architecture Walkthrough

Step 2

Neighbourhood aggregation (layer k):

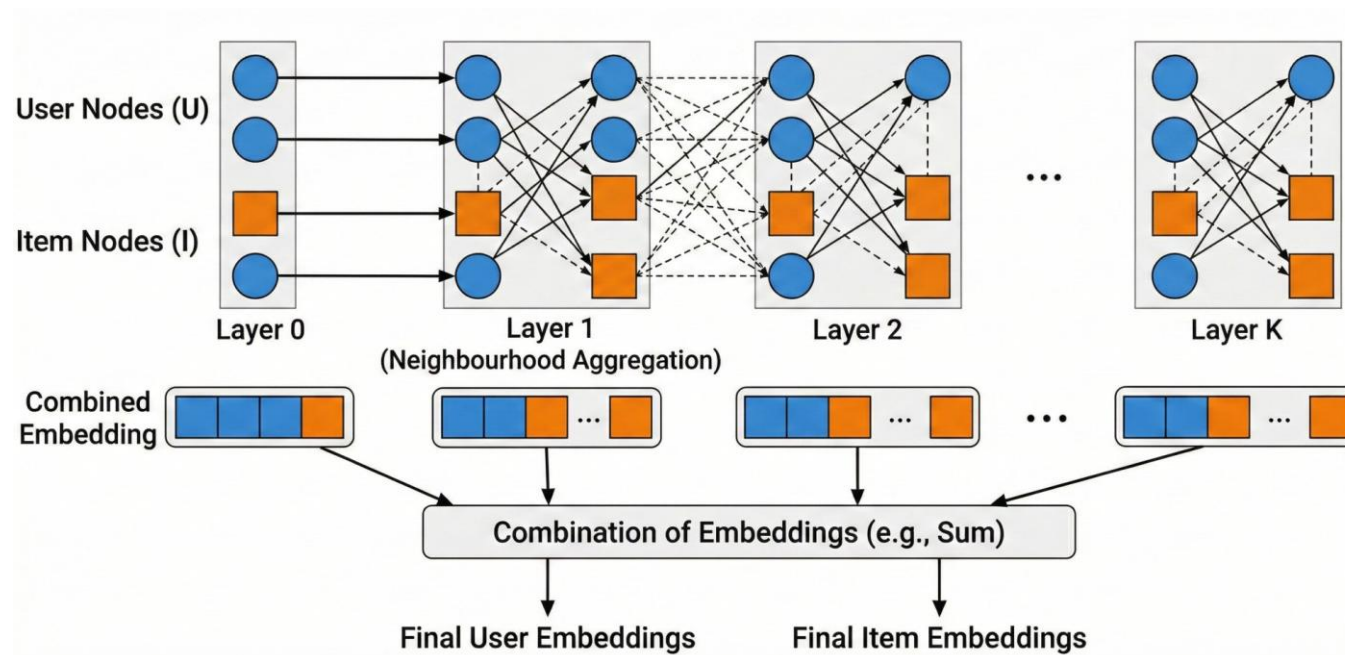
$$h_v^{(k)} = \sum_{u \in \mathcal{N}(v)} \tilde{A}_{vu} h_u^{(k-1)}$$

Layer combination:

$$z_v = \sum_{k=0}^K h_v^{(k)}$$

Prediction:

$$\hat{r}_{ui} = z_u^T z_i$$



Key results findings

- Best of all baselines
- Biggest gain in recall
- Captures multi-hop patterns invisible to matrix methods
- Limitation: link-prediction, not Top-N

Method	Precision	Recall
UCBF	0.7423	0.8015
IBCF	0.7689	0.8247
MF	0.8012	0.8619
SVD	0.8294	0.8896
SVD++	0.8627	0.9138
LightGCN-CF	0.9101	0.9625

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Objectives

Methodology

Results

Findings

Answers to the questions and objectives

- Higher recall indicates that multi-hop propagation captures relevant interactions missed by similarity- and latent-factor methods.
- The progression from UBCF to GNN-CF suggests that richer relational modelling improves link prediction.
- However, the results do not yet establish explainability, temporal adaptation, or full-catalog top-N performance.

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Objectives

Methodology

Results

Findings

Key findings

- **Graph structure improves prediction:** Interaction connectivity adds value beyond matrix-based modelling.
- **Higher-order neighborhoods improve recall:** Indirect signals recover interactions missed by direct similarity.
- **Simple architectures are sufficient:** Complex nonlinear GNN components are unnecessary for a strong baseline.
- **Accuracy is not enough:** The model remains static and opaque.
- **Evaluation scope must remain clear:** the contribution addresses link prediction, not top-N ranking.

Limitations

- The model closely follows LightGCN and is not a fundamentally new GNN architecture.
- Evaluation is limited to MovieLens-100k under a link-prediction protocol.
- No ablation isolates propagation depth or layer aggregation.
- Interactions are uniformly weighted, static, and not explicitly explainable.
- The contribution describe the experiment as link prediction, not top-N recommendation.

Takeaways

- Graph propagation is an effective representation mechanism,
- Graph propagation has a new limitation: All historical interactions influence the graph model implicitly and uniformly.
- The next contribution is therefore motivated by two questions:
 - Which historical interactions actually drive a recommendation?
 - Can their influence be adjusted during recommendation generation rather than remaining fixed?

Takeaway – Contribution I

Motivation to use Shapley values to estimate interaction-level contributions, convert them into adaptive weights, and produce human-understandable evidence for each recommendation.

5

Contribution II

Dynamic Recommender Systems with Real-time Shapley Value-based Contribution Adjustment

Research Gap

- Most recommender systems make two simplifying assumptions:
 - User preferences remain stable
 - All historical interactions have comparable relevance to the current recommendation.
- Adaptive methods often rely on heuristic weights without explaining their importance.
- Post-hoc explanations describe predictions without influencing them.

Research Gap

The absence of a unified, model-agnostic mechanism that quantifies interaction importance, uses it to adapt recommendations, and exposes it as an explanation.

[26] O. A. S. Ibrahim, E. M. G. Younis, E. A. Mohamed, and W. N. Ismail, "Revisiting Recommender Systems: An Investigative Survey," *Neural Computing and Applications*, vol. 37, pp. 2145–2173, 2025.

[27] P. Mateos and A. Bellogin, "A Systematic Literature Review of Recent Advances on Context-Aware Recommender Systems," *Artificial Intelligence Review*, vol. 58, Art. no. 20, 2025.

[28] X. Fan and Y. Ji, "A Dynamic Preference Recommendation Model Based on Spatiotemporal Knowledge Graphs," *Complex & Intelligent Systems*, vol. 11, Art. no. 46, 2025.

Research Questions → Objectives

ID	Research Question	Objective
RQ2.1	Can Shapley values quantify each interaction’s contribution to a recommendation?	Model interactions as cooperative-game players using recommendation score as the value function.
RQ2.2	Does Shapley-based reweighting improve top-10 ranking across models and datasets?	Apply adaptive weights to UBCF, IBCF, MF, and NCF on MovieLens-1M and Amazon Reviews.
RQ2.3	Can contribution scores both adjust and explain recommendations?	Use Shapley values to reweight recommendations and identify influential interactions.
RQ2.4	Do performance gains justify the computational cost?	Compare runtime, memory use, and cost-benefit trade-offs.
RQ2.5	Does the study demonstrate long-term preference adaptation?	Distinguish offline reweighting from streaming adaptation and limit the claim accordingly.

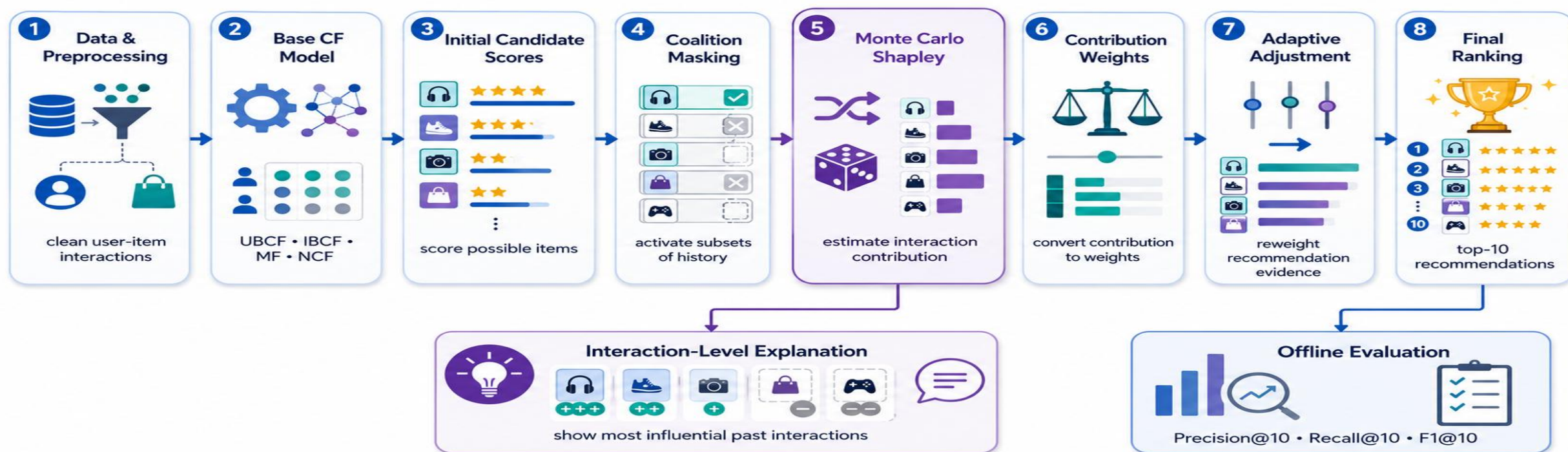
Objectives

Methodology

Results

Findings

Architecture Walkthrough



Architecture Walkthrough

Let $\mathcal{I}_u = \{(u, i_1), \dots, (u, i_n)\}$ be user u 's historical interactions. For candidate item j and coalition $\mathcal{C} \subseteq \mathcal{I}_u$

$$v(\mathcal{C}) = \hat{r}_{uj}(\mathcal{C})$$

The Shapley value of interaction e is its expected marginal contribution over all coalitions:

$$\phi_e = \sum_{\mathcal{C} \subseteq \mathcal{I}_u \setminus \{e\}} \frac{|\mathcal{C}|! (|\mathcal{I}_u| - |\mathcal{C}| - 1)!}{|\mathcal{I}_u|!} [v(\mathcal{C} \cup \{e\}) - v(\mathcal{C})]$$

Shapley scores become adaptive interaction weights:

$$w(u, j) = 1 + \alpha \cdot \phi(u, j)$$

Applied directly inside the scoring rule of the base model, e.g. for user-based CF:

$$\hat{r}_{ui}^{adapt} = \bar{r}_u + \frac{\sum_{v \in N_u} w(v, i) s(u, v) (r_{vi} - \bar{r}_v)}{\sum_{v \in N_u} |s(u, v)|}$$

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Objectives

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Evaluation protocol

- **Datasets:** MovieLens-1M and Amazon Product Reviews.
- **Task:** sampled-candidate top-10 recommendation.
- **Candidate set:** held-out positives plus 99 sampled unobserved items per user.
- **Metrics:** Precision@10, Recall@10, and F1@10.
- **Evidence limitation:** the main values are point estimates; complete multi-seed variance was not available.

Key results findings - Movielens 1M

- ASV improves every baseline, confirming that contribution-aware weighting is broadly effective.
- ASV-MF achieves the best overall F1 score at 0.840, with a 9.4% improvement.
- ASV-NCF shows the smallest gain, suggesting limited improvement when the baseline is already strong.

Model	Precision@10	Recall@10	F1@10
UBCF	0.750	0.650	0.696
ASV-UBCF	0.820	0.709	0.760
IBCF	0.780	0.700	0.738
ASV-IBCF	0.851	0.763	0.805
MF	0.810	0.730	0.768
ASV-MF	0.886	0.798	0.840
NCF	0.850	0.780	0.813
ASV-NCF	0.873	0.796	0.832

Key results findings – Amazon Products

- ASV-IBCF achieves the largest F1 improvement at 12.0%.
- ASV-NCF also shows a strong gain, improving F1 by 10.9%.
- ASV-MF records the highest F1 score, but its improvement is only 0.95%.

Model	Precision@10	Recall@10	F1@10
UBCF	0.698	0.673	0.685
ASV-UBCF	0.768	0.740	0.753
IBCF	0.705	0.696	0.700
ASV-IBCF	0.790	0.779	0.784
MF	0.886	0.800	0.841
ASV-MF	0.898	0.808	0.849
NCF	0.742	0.715	0.728
ASV-NCF	0.841	0.776	0.807

Computational overhead

- ASV-IBCF has the highest overhead.
- ASV-NCF has the lowest training overhead.
- All variants increase latency and memory.

Enhanced Method	Training-Time Overhead	Prediction-Time Change	Memory Change
ASV-UBCF	+60%	8.7 → 14.2 ms	145 → 189 MB
ASV-IBCF	+83%	7.4 → 13.8 ms	138 → 185 MB
ASV-MF	+39%	12.1 → 18.9 ms	267 → 341 MB
ASV-NCF	+30%	15.4 → 24.7 ms	412 → 523 MB

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Answers to the questions and objectives

- Each interaction receives a recommendation-specific Shapley contribution.
- Across both datasets, all ASV variants improve their baselines, with F1 gains from 0.95% to 12.0%.
- The same scores support both reweighting and interaction-level explanation.
- Overhead can be substantial: ASV-MF performs best on MovieLens, ASV-NCF on Amazon, while ASV-MF offers weak value on Amazon.
- The results show offline contribution-aware adjustment, not long-term streaming adaptation.

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Key findings

- Broadly reusable: Improves neighborhood, latent-factor, and neural models.
- Dual purpose: Shapley values support both weighting and explanation.
- Uneven gains: Strong baselines may have limited headroom, as with ASV-MF on Amazon.
- Efficiency cost: Monte Carlo reduces complexity but increases training, prediction, and memory use.
- Limited adaptivity: Supports inference-time or precomputed reweighting, not continuous production learning.
- Different integration: UBCF/IBCF adjust scores directly; MF/NCF use offline enhancement.

Limitations

- **High computation:** Shapley estimation is costly for long histories.
- **Clean input bias:** Implicit positives are less noisy than real production signals.
- **No drift testing:** Controlled preference changes were not evaluated.
- **No live validation:** No A/B test or dedicated user study was conducted.
- **Limited statistics:** Multi-seed variance and paired significance tests are missing.
- **Limited model scope:** Deep graph recommenders were not tested.
- **Sampled ranking bias:** Results may overstate full-catalog performance.

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Takeaways

- **Current limitation:** The current contribution models flat user-item structures in Euclidean space.
- **Hierarchical domains:** Movies map to genres, while products belong to nested categories.
- **Preference propagation:** Item preferences may influence related categories and their ancestors.

Takeaway – Contribution II

We Must address Euclidean distortion and flat explanations by modeling user-item-category hierarchies in non-Euclidean space, extending Shapley attribution, generating multi-level explanations, and evaluating their faithfulness and usefulness.



Contribution III

Graph Neural Networks with Shapley-Value Explanations for Hierarchical Recommendation Systems

Research Gap

- Products are organized into categories and subcategories
- Movies belong to genres
- Preferences can exist at item, category, and higher abstraction levels
- Parent–child relations create long-range dependencies.

Research Gap

Hyperbolic models are more expressive for hierarchical data but harder to interpret. Existing Shapley-based explainers are commonly designed around Euclidean scoring and do not automatically preserve consistency with hyperbolic distance.

[29] T. Markchom, H. Liang, and J. Ferryman, "Review of Explainable Graph-Based Recommender Systems," arXiv:2408.00166, 2024.

[30] X. He, "Graph Neural Networks in Recommender Systems," Applied and Computational Engineering, vol. 79, pp. 184–191, 2024.

[31] X. Yang et al., "HgFormer: Hyperbolic Graph Transformer for Collaborative Filtering," Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.

[32] Y. Zhang et al., "Hyperbolic Graph Representation Learning: Methods, Applications and Challenges—A Survey," Neurocomputing, vol. 652, Art. no. 131044, 2025.

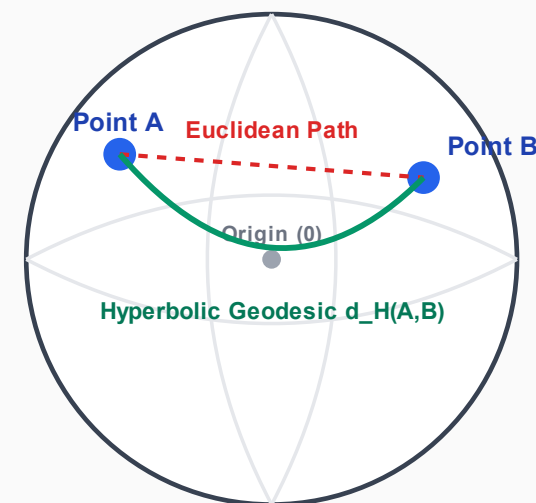
Research Questions → Objectives

ID	Research Question	Objective
RQ3.1	Does hyperbolic learning improve accuracy for hierarchical user–item relations?	Compare the full model with Euclidean, flattened, LightGCN, and Hyperbolic GCN baselines.
RQ3.2	Can Shapley explanations be added without reducing performance?	Compute hyperbolic Shapley contributions and compare with a non-Shapley variant.
RQ3.3	Do explanations reflect hierarchy and user preferences?	Test hierarchical relevance, faithfulness, user judgments, and preference alignment.

Hyperbolic Space

- A curved, negatively-curved space. Distances grow exponentially outward, matching how a tree or hierarchy grows.
- First linked to real networks used for representation learning of hierarchies and then extended to graph neural networks.
- Movies belong to genres, products to categories, hyperbolic space represents this hierarchy naturally and efficiently. This motivates Contribution III.

A. Curved Space & Geodesics



- **Negative Curvature ($c < 0$):** Bends geodesics inward.
- Metric distance: $d_c(x, y) = 2 \cdot \text{artanh}(\|x \oplus_c y\|)$
- **Shortest paths curve toward center origin naturally.**
- Boundary rim acts as infinite mathematical horizon.

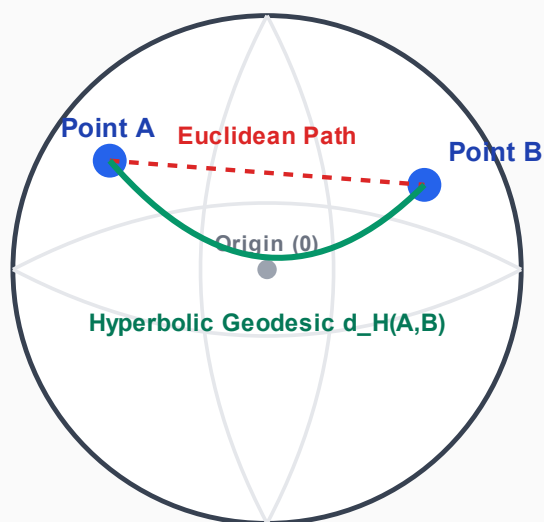
Objectives

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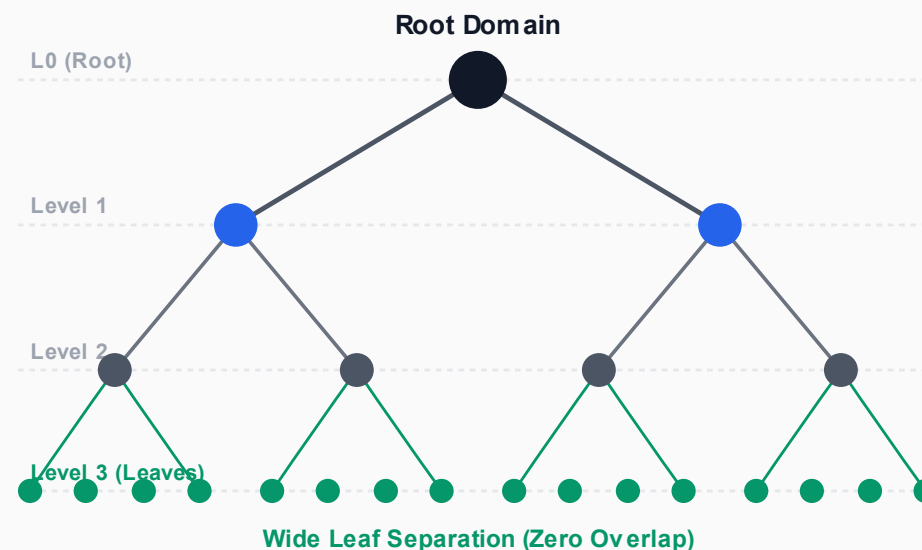
Results

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A. Curved Space & Geodesics



- **Negative Curvature ($c < 0$):** Bends geodesics inward.
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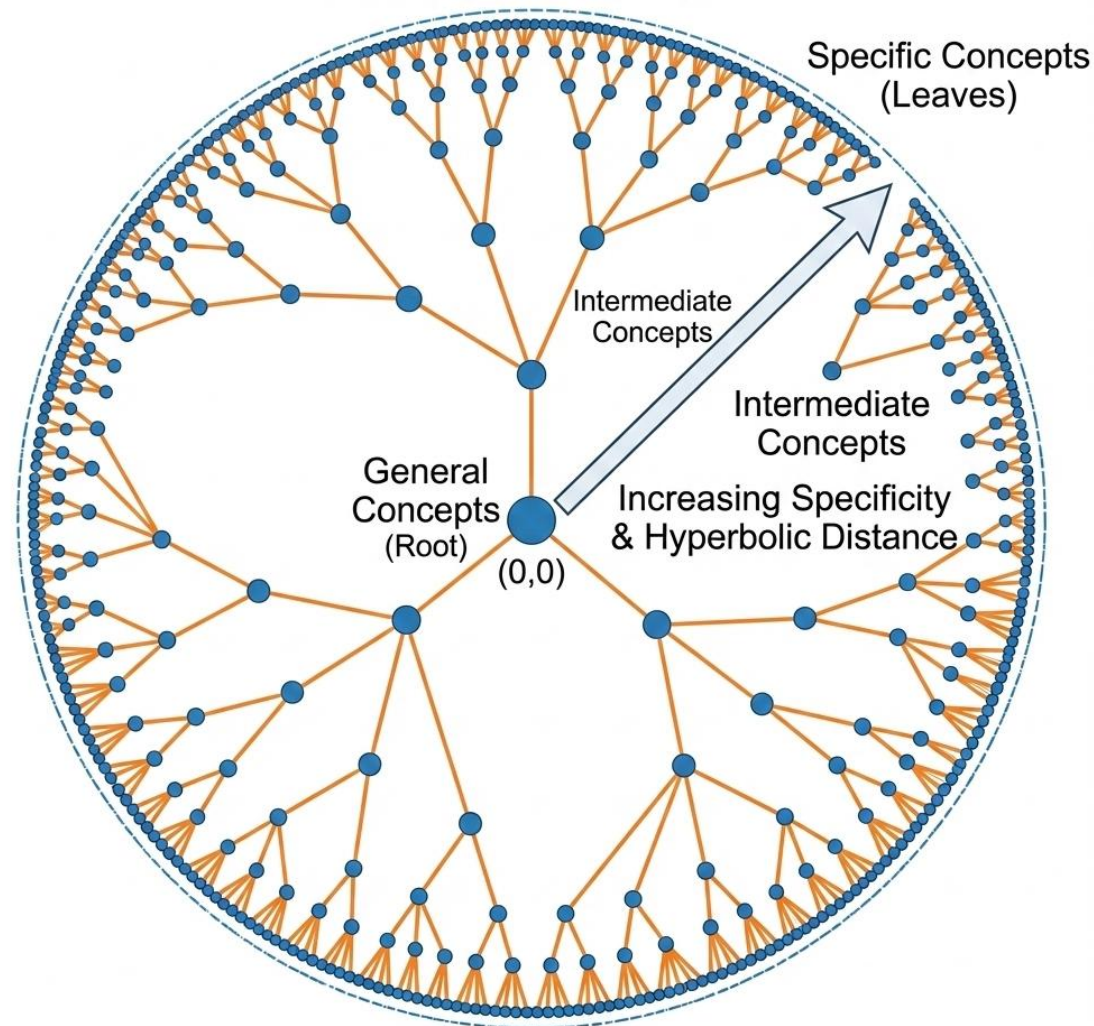
B. Hierarchical Representation in H^2 

Low Distortion in Deep Hierarchies

- Surface area expands exponentially $O(e^r)$ with radius.
- Exponential growth matches 2^L tree capacity precisely.
- **Result: Deep categories & items remain distinct without distortion.**

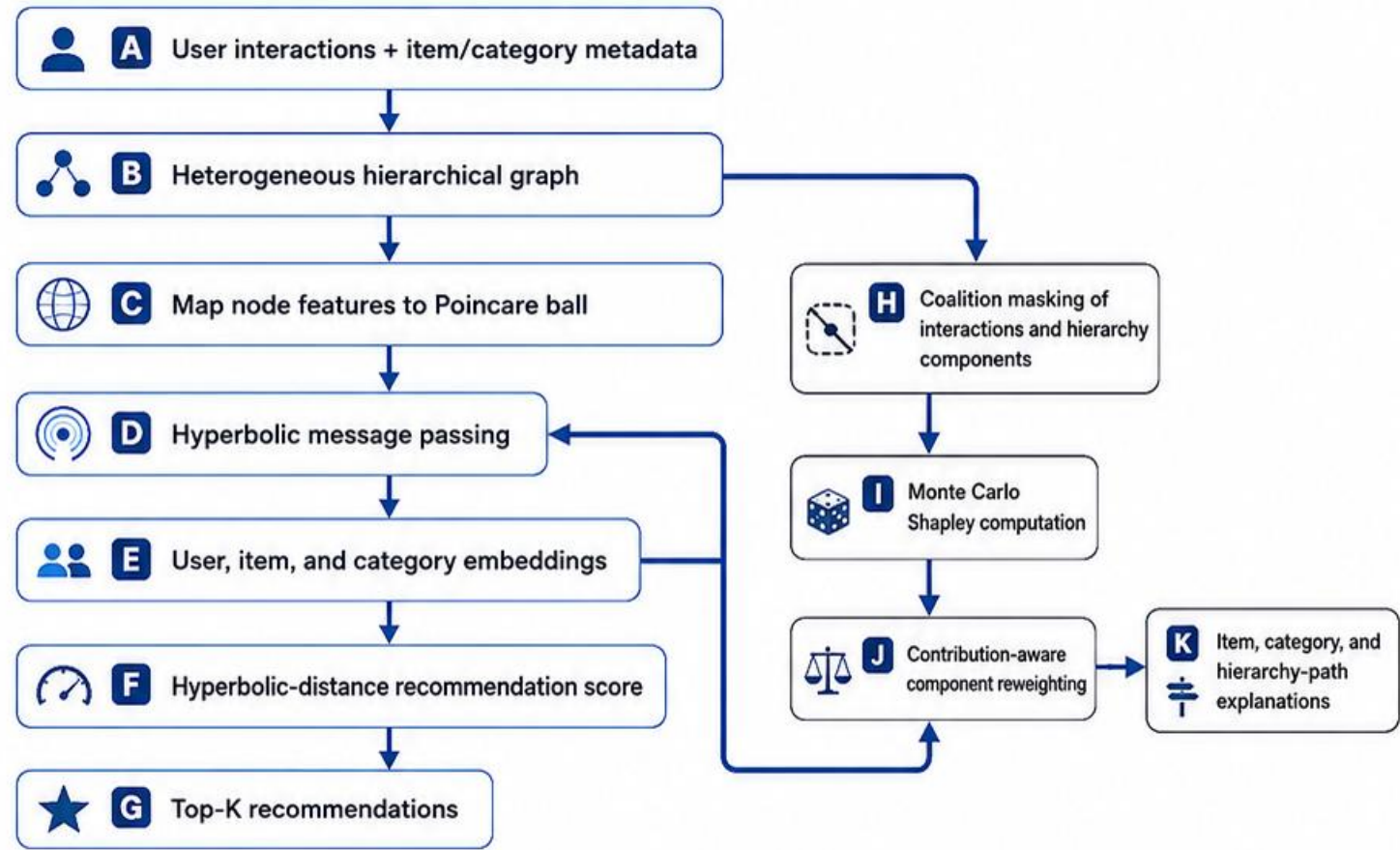
The Poincare Model

- A practical model for computing in hyperbolic space. Points sit inside a ball; the edge represents infinity, and the maths stays compatible with gradient-based training.
- It is used it for word hierarchies like WordNet, hyperbolic GNNs; recent 2024–2025 work applies it directly to hierarchical recommendation.
- It is the best-supported hyperbolic model for deep learning, integrates directly into GNN training, and has proven successful in prior hierarchical recommendation work.



[34] M. Nickel and D. Kiela, "Poincaré Embeddings for Learning Hierarchical Representations," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2017, pp. 6338–6347.

Architecture Walkthrough



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Evaluation protocol

- **Datasets:** MovieLens-1M and Amazon Product Reviews.
- **Task:** sampled-candidate top-k ranking.
- **Candidate set:** held-out positives plus 99 sampled unobserved items per user.
- **Metrics:** Precision@10, Recall@10, and F1@10.
- **Evidence limitation:** Ablation, Runtime, Perturbation faithfulness, and user study.

Key results findings - Movielens 1M

- Hyperbolic GNN + Shapley achieves the best results across all metrics.
- Hyperbolic GCN outperforms LightGCN, showing the value of hierarchical geometry.
- Recall@10 shows the largest improvement, indicating better retrieval of relevant items.

Method	Precision@10	Recall@10	F1@10
LightGCN	0.763	0.703	0.732
Hyperbolic GCN	0.795	0.742	0.768
LightGCN + GNNShap	0.774	0.715	0.743
LightGCN + MAGE	0.783	0.731	0.756
HGNN-SV	0.822	0.785	0.803

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Ablation study - Movielens 1M

Variant	Precision@10	Recall@10	F1@10	F1 drop
Full model	0.822	0.785	0.803	—
Without Shapley	0.816	0.774	0.794	-0.009
Without hyperbolic geometry	0.792	0.749	0.770	-0.033
Without hierarchy	0.758	0.723	0.740	-0.063

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Ablation study – Amazon Products

Variant	Precision@10	Recall@10	F1@10	F1 drop
Full model	0.785	0.730	0.756	—
Without Shapley	0.777	0.716	0.745	-0.011
Without hyperbolic geometry	0.753	0.692	0.721	-0.035
Without hierarchy	0.719	0.663	0.690	-0.066

Computational overhead

- Training is 33–38% slower than Hyperbolic GCN.
- Inference is 41–50% slower across the two datasets.
- The overhead is manageable experimentally, but production readiness is not yet established.

Method	MovieLens train / infer (s)	Amazon train / infer (s)
LightGCN	8.5 / 1.2	22.3 / 3.1
Hyperbolic GCN	11.7 / 1.6	27.9 / 4.4
LightGCN + GNNShap	14.2 / 2.3	35.1 / 5.8
Full model	15.6 / 2.4	38.5 / 6.2

Answers to the questions and objectives

- **Supported within protocol:** GNN-CF achieved the highest precision and recall.
- **Partially supported:** Recall gains suggest benefits from multi-hop collaborative signals, but propagation depth was not isolated.
- **Supported:** A simple LightGCN-style model performed best without complex transformations.
- **Clearly established:** The model uses uniform edge weights, cannot reweight interactions, and lacks local explanations.

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Key findings

- Graph structure improves edge-prediction beyond matrix methods.
- Higher-order neighborhoods mainly improve recall.
- Simple LightGCN-style architecture is sufficient as a strong baseline.
- Predictive accuracy does not address static weighting or explainability.
- Results should be framed as link prediction, not top-N ranking.

Limitations

- The model closely resembles LightGCN and is not a fundamentally new GNN architecture.
- Evaluation is limited to MovieLens-1M and link prediction.
- No ablation isolates propagation depth or layer aggregation.
- All interactions are weighted uniformly.
- The model provides no local explanations.
- Preference adaptation is not validated.
- Claims should follow the link-prediction experiment and not imply top-N ranking.

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Takeaways

- Graph propagation is effective, but all historical interactions are treated uniformly.
- Shapley values identify which interactions drive each recommendation.
- Contribution scores enable adaptive weighting and understandable explanations.

Takeaway – Contribution III

The contribution moves beyond static graph propagation by measuring interaction importance. It uses these contributions both to improve recommendation generation and to explain the result.

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Conclusion & Perspectives

Contribution	Main Idea	Main Achievement	Key Finding
C1	Explainable graph-based recommendation using Shapley values	Introduced faithful explanations for GNN-based recommender systems by quantifying the contribution of historical user-item interactions.	Explainability improves transparency while preserving recommendation performance.
C2	Adaptive recommendation through contribution-aware inference	Leveraged Shapley contributions to dynamically reweight historical interactions during inference.	Adaptive weighting improves recommendation robustness and personalization.
C3	Hyperbolic graph neural networks for recommendation	Modeled hierarchical user-item relationships in hyperbolic space to better capture latent structures.	Hyperbolic representations outperform Euclidean embeddings for hierarchical recommendation.

Takeaway – Thesis

Overall, this thesis progressively integrates explainability, adaptivity, and hyperbolic representation learning into a unified framework for accurate, transparent, and robust graph-based recommender systems.

Published Research Papers

No.	Title	Journal	Status
I	Dynamic Recommender Systems with Real-time Shapley Value-based Contribution Adjustment	International Journal of Intelligent Engineering and Systems	Published, 2025
II	Graph Neural Networks with Shapley-Value Explanations for Hierarchical Recommendation Systems	International Journal of Advanced Computer Science and Applications	Published, 2025
III	Shapley Values for Explaining Adaptive Recommendations in Temporal Hyperbolic Graph Neural Networks	Discover Artificial Intelligence	Under review
IV	Toward Temporal Stability in Explainable AI: A Framework for Measuring Explanation Drift	Results in Engineering	Under review

Published Research Papers

No.	Title	Conference	Status
V	A Collaborative Filtering Movies Recommendation System based on Graph Neural Network	ANT / EDI40	Published, 2023
VI	A Hybrid Machine Learning Method for Movies Recommendation	BDIoT	Published, 2022

- Limited domain evaluation: Experiments were conducted on benchmark datasets (MovieLens and Amazon) validation on additional real-world domains remains necessary.
- Scalability of Shapley-based explanations: Although optimized, computing Shapley values remains computationally expensive for very large-scale recommender systems.
- Static graph assumption: The proposed models operate on static interaction graphs and do not explicitly model continuously evolving user–item relationships.
- Limited multimodal information: The framework primarily exploits user–item interaction data, without integrating textual, visual, or knowledge graph information.
- Hyperbolic architecture scope: The hyperbolic approach was evaluated on a specific HGNN architecture; its generalization to other graph neural network models remains to be investigated.
- Offline evaluation: Performance was assessed through offline experiments and user studies, without validation in an online production environment (e.g., A/B testing).

- Dynamic Graph Recommender Systems: Extend the proposed framework to temporal and continuously evolving user-item interaction graphs.
- Scalable Explainable Recommendation: Develop efficient approximations of Shapley values to support real-time explanations in large-scale recommender systems.
- Multimodal Recommendation: Integrate textual, visual, and knowledge graph information to improve both recommendation accuracy and explanation quality.
- Advanced Hyperbolic Graph Learning: Investigate more expressive hyperbolic architectures (e.g., Hyperbolic Graph Transformers) and mixed-curvature embedding spaces for complex hierarchical data.
- Online and Real-World Deployment: Validate the proposed methods through online A/B testing and deployment in industrial recommender systems to assess long-term user satisfaction, trust, and business impact.

Thesis Answer

This thesis demonstrates that explainability should not be considered only as a post-hoc interpretation tool, but as a fundamental component for building accurate, adaptive, and trustworthy graph-based recommender systems.

Key Outcomes

- Unified framework combining GNNs, Shapley values, adaptive inference, and hyperbolic geometry.
- Interpretable recommendations based on interaction-level contribution analysis.
- Adaptive personalization through contribution-aware reweighting.
- Improved hierarchical modeling with hyperbolic graph neural networks.
- Validated performance on MovieLens and Amazon datasets.

Questions & Discussion

Thank you for your attention.
I welcome your questions and comments.

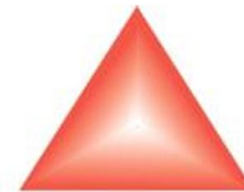


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المملكة المغربية
وزارة التعليم العالي
و البحث العلمي و الابتكار



المدرسة الوطنية العليا للمعلوماتية و تحليل النظم
École Nationale Supérieure d'Informatique
et d'Analyse des Systèmes

Doctoral Studies Center in Information and Engineering Sciences and Technologies (ST2I)

Explainable Artificial Intelligence (XAI) for Decision-Making in Graph-Based Recommendation Systems

PhD Viva Presented by: Redwane NESMAOUI

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